

Pressure Distributions on Sloped Gradients

By

Ethan Goldman, Tyler Mackowski, Carolyn Thompson, Madanna Vue

Final Project Report

Mechanical Engineering 70 - Experiments / Instruments

Tufts University

Medford, Massachusetts

May 3, 2024

Table of Contents

Executive Summary.....	2
1. Introduction.....	2
2. Methods.....	3
2.1 Fundamental Theory.....	3
2.2 Physical Setup and Software Summary.....	4
2.3 Sensor Calibration.....	5
3. Results.....	5
4. Discussion.....	7
5. Conclusion and Future Work.....	8
References.....	9
Milestone 1: Project Proposals.....	10
Milestone 2: Experimental Plan.....	11
Milestone 3 (Background Section):.....	12
Milestone 4: Documentation of Constructed Experimental Setup.....	13
Milestone 5: Software Development.....	15
Milestone 6: Foot Pressure Gradient Sensor Preliminary Data.....	16

Executive Summary

In this project, our goal was to create a pressure gradient sensor using load cell technology and LabVIEW to analyze foot pressure data and produce a foot pressure gradient. The data found using the load sensors can then be used to make recommendations about foot comfort in shoes. Load cells are sensory devices that measure a load when applied through a Wheatstone bridge circuit. This technology is helpful as it is inexpensive and accessible to engineers and can measure large loads without overloading the sensors. An array of load cells was created and placed into a shoe, which was then used in a stationary position by the test subject on an incline, decline, and flat surface. Over a span of 45-60 seconds, data was collected on the force at each of the six sensors while the user stood still, shoulder-width apart. This data was then put through LabVIEW code, in which a pressure map of the test subject's foot was generated to show how much force was applied to different foot positions, from toe to heel. The data concluded that at a 6.6° incline, the pressure distribution was greatest in the heel; at a 6.6° decline, more pressure was found in the front of the foot and toe, and in a neutral, flat position, the pressure was relatively evenly distributed with more found in the medial midfoot. The data from this study shows the accuracy of the pressure gradient sensor using the load cells. These conclusions also led to the recommendation for better shoe inserts, in which ones with increased arch support would allow for better overall pressure distribution and, therefore, comfort for everyday use.

1. Introduction

The study focuses on applying load cells to create an inexpensive yet accurate pressure gradient sensor to examine the pressure distributions of a foot standing on an uphill and downhill incline. The findings from this experiment will explore the application of load cell technology. Our feet are essential to our everyday functioning and endure much pressure from our bodies. Complications like Plantar Fasciitis mainly occur from foot overuse (Hoffman, 2024). Additionally, people with pre-existing conditions like flat feet suffer from inadequate foot support in everyday shoes. In our study, we created a foot pressure sensor using load cells that produce pressure gradient heat maps that will help analyze foot pressure distribution and aid in making recommendations to improve foot comfort at different inclines. Currently, a significant amount of data exists about foot pressures experienced on uphill and downhill slopes, along with different walking/running conditions. Existing data on foot pressures experienced during uphill and downhill walking determined that “the peak pressure on the [medial midfoot] during downhill slope walking was lower than that during ground and uphill walking, whereas the peak pressure on the [medial midfoot] during uphill slope walking was similar to that during ground walking at each walking speed” (Koo et al.). However, while the Koo et al. study was performed on walking subjects, our study was performed on stationary subjects, so while the data from Koo et al. may be helpful to validate our data, variation in the data between the two studies does not necessarily invalidate our data. Our study suggests a new way to build an inexpensive yet accurate pressure gradient sensor. It uses that sensor to collect preliminary data that contributes to a new understanding of stationary pressure distributions at different inclines.

2. Methods

This project aims to obtain data from pressure sensors to understand the pressure distributions at the foot when standing at different inclines when someone is standing on the sensors. In this study, we created an array of sensors at the base of a women's size 5 Converse Chuck Taylor shoe. The sensors read data through a strain gauge and are read from analog pins from an ESP32 board. This data is then communicated through serial to LabVIEW, which is analyzed to produce a pressure heat map.

2.1 Fundamental Theory



FIGURE 1. A Free Body Diagram of the Converse shoe in the flat (neutral) position. The CoM is expected to be over the mid-sole of the shoe, contributing to a larger normal force distribution in the mid-sole.

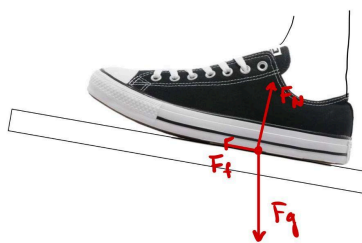


FIGURE 2. A Free Body Diagram of the Converse shoe in an inclined position. The CoM is expected to be closer to the heel. The normal force is directed towards the heel, while the frictional force is colinear with the slope's incline towards the toes. A larger force distribution in the heel of the sole is expected compared to the neutral condition.

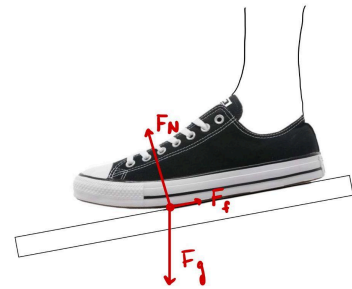


FIGURE 3. A Free Body Diagram of the Converse shoe in a declined position. The CoM is expected to be closer to the front of the foot. The normal force is directed toward the toes while the frictional force is colinear with the slope inclined of the slope, towards the heel. It is expected to have a larger force distribution in the front of the sole compared to the neutral condition.

To build our model, we created an array of load cells that would take force data at different pressure points at the foot when standing. The positions of the sensor of the array were referenced from various sources. The main criteria for the sensors were that they could withstand a large load (≤ 50 kg/ 33.7 lbs). This criterion was important because we wanted to ensure that the sensor would not be overloaded when someone was standing on the sensor array. To prevent sensory overload, we also decided to only take the foot pressure from one foot while the subject stands on both feet in a neutral position. By doing this, we take off half the body weight applied to the sensor and reduce the load applied to the sensor. Our final model used 50 kg load sensors which use a strain gauge to measure a force when applied.

2.2 Physical Setup and Software Summary

An array of sensors was built in the base of a Women's size 5 Converse Shoe to collect pressure data. The load cell sensors were arranged in specific pressure points of the foot (Bartlett, 1997), mainly at the big toe, the ball of the foot, mid-foot, and heel as seen in Figure 4.

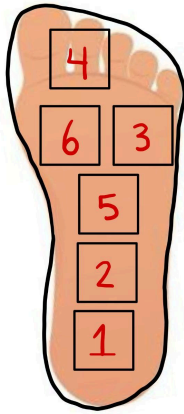


FIGURE 4. Depiction of the pressure sensor array in the created in the insole of the Chuck Taylor Converse shoe. Pressure sensors are strategically placed to measure the force distribution in the three areas of interest: the heel, mid-foot, and toes.

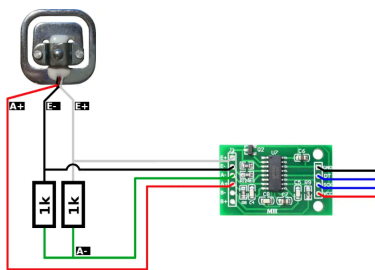


FIGURE 5. A wiring diagram of the 50 kg load cell and a HX711 amplifier. The schematic represents a Wheatstone Bridge, where the load cell is half of the bridge, and the resistors complete the bridge. A HX711 is required to acquire data because it acts to amplify signals from cells and report them to another microcontroller. The HX711 is a precision 24-bit analog to digital converter (ADC).



FIGURE 6. Image of load cell sensors laid into the Chuck Taylor Converse. Sensors were mounted using 3D printed mounts inlaid into a sheet of foam. The array of sensors is consistent with the schematic.

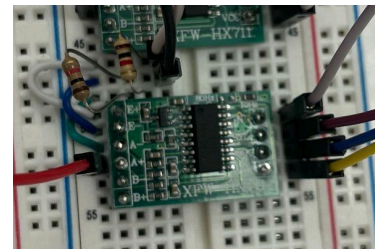


FIGURE 7. Image of the wiring from the sensors to the HX711 (on the left side) and from the HX711 to the ESP-32 (on the right side). Two 1,000 Ohms resistors create the Wheatstone Bridge in the breadboard. Every load cell requires its own HX711 to amplify the signal before being converted to the ESP-32.

2.3 Sensor Calibration

Calibration						
Sensor	Average Offset Value	1 kg	3 kg	5 kg	Slope	B-Value
1	-696,000	-22,000	-96,000	-172,000	-2.67E+05	-0.4224
2	1,357,000	-51,000	-134,000	-214,000	-2.45E+05	-0.2634
3	-2,743,000	-58,000	-140,000	-220,000	-2.47E+05	-0.4402
4	-2,915,000	-48,000	-130,000	-210,000	-2.47E+05	-0.1933
5	800,000	-27,000	-107,000	-190,000	-2.45E+05	-0.35
6	-4,776,000	-55,000	-140,000	-225,000	-2.35E+05	-0.2941

Table 1: Calibration data for each of the six load cell sensors. Arduino IDE was used to plot the fluctuating range of values when no load was applied to the pressure sensors to zero the noise coming from each sensor. The data was plotted until convergence was reached with the average offset value. This number was used to zero each sensor before the load calibration occurred. To solve for the load conversion for each sensor, the shoe was placed on a scale to determine the applied force, and a force was applied to a singular load cell at a time. Trials of 1, 3, and 5 kg were applied to each sensor, and data was recorded in the plot above. These values were analyzed using a MatLab script to create a line of best fit, calculating the slope and b-values for the linear computation of load using an additional Arduino code.

3. Results

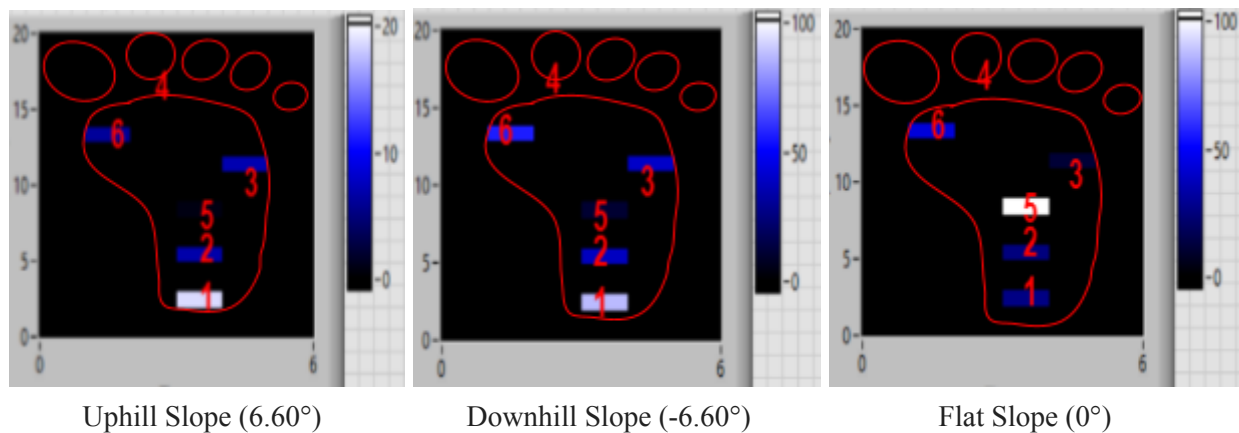


FIGURE 8. Observed foot pressure gradients on uphill, downhill, and flat slopes. Absolute force values are known to not be accurate, but relative values between the sensors are likely accurate. Data from sensor 4 was not plotted correctly by LabVIEW and thus does not appear on the heatmap. Data was collected for approximately 30 seconds. Maximum relative pressure in the uphill and downhill conditions was observed in the heel, and for the flat slope condition was observed in the midfoot.

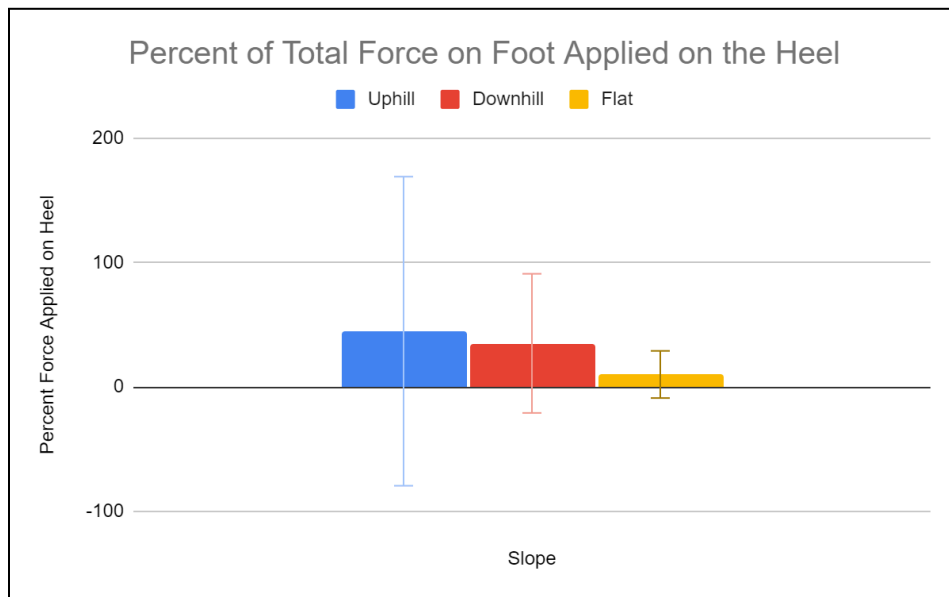


FIGURE 9. Percent of total force applied in the heel of the foot for the uphill, downhill, and flat surface conditions plotted with a 95% confidence interval. The heel of the foot experienced the most relative force in the uphill condition and the least in the flat condition. However, the data appears to be non-conclusive because of the overall of the confidence intervals from all three conditions.

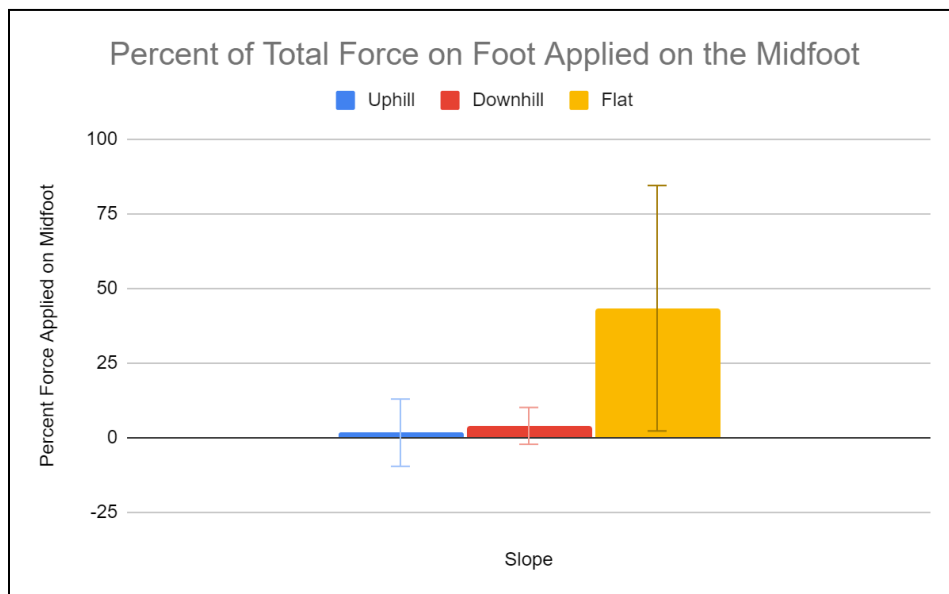


FIGURE 10. Percent of total force applied in the midfoot region for the uphill, downhill, and flat surface conditions plotted with a 95% confidence interval. The midfoot experienced a much larger relative force in the flat condition than in both the uphill and downhill conditions. However, the data appears to be non-conclusive because of the overall of the confidence intervals from all three conditions.

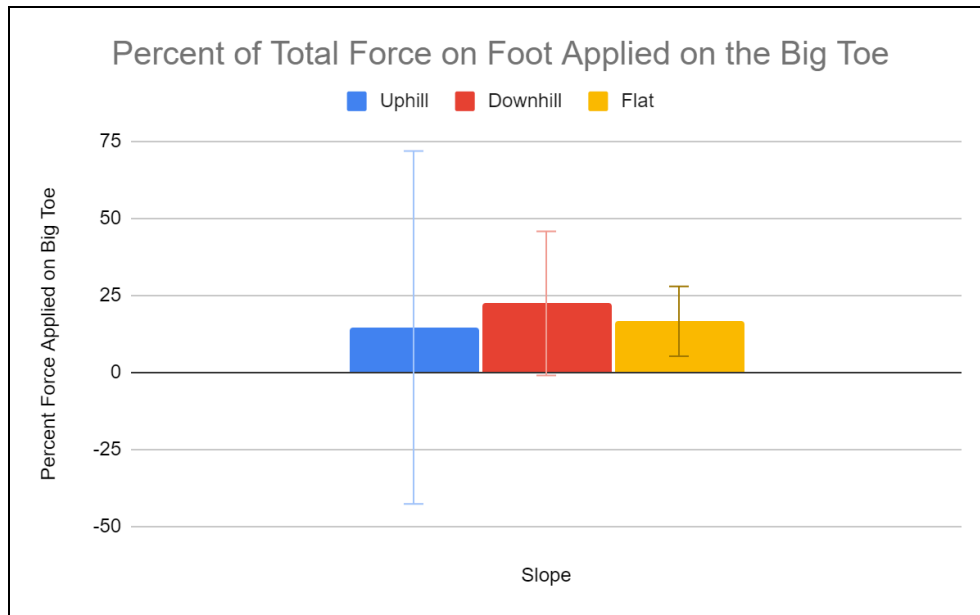


FIGURE 11. Percent of total force applied in the big toe region for the uphill, downhill, and flat surface conditions plotted with a 95% confidence interval. The big toe experienced the most relative force in the downhill condition and the least in the uphill condition. However, the data appears to be non-conclusive because of the overall confidence intervals from all three conditions.

4. Discussion

Preliminary tests of the sensor that we made give positive results for sensor functionality. Our free body diagrams of the foot in flat, uphill, and downhill conditions predicted that we would see the most force in the heel in the uphill condition, the most force in the midfoot in the flat condition, and the most force in the front of the foot in the downhill condition. Ignoring the very large confidence intervals (see paragraph two for a discussion of the confidence intervals), these predictions are corroborated by the data that we took. Figure 9 shows that the heel experienced the most relative force in the uphill condition. Unexpectedly, there was more relative force in the heel in the downhill condition than in the flat condition – this result may be a result of miscalibration of the sensor in the heel because, as shown in Figure 9, the heel accounted for over 30% of the total force of the foot in both uphill and downhill conditions which is an unexpected result. Figure 10 shows that the midfoot experienced the most relative force in flat conditions, as expected. Figure 11 shows that the front of the foot experienced the most force in the downhill condition, as expected.

The confidence intervals for all of the data we took were very large – so large that our data is inconclusive. This result is surprising because while we did not standardize the amount of time each test ran for, the test for each condition was run for at least 30 seconds, which we thought would be enough time to get a much smaller confidence interval. This problem could be solved with a wide range of solutions. As mentioned before, one possibility is simply standardizing and increasing the amount of time each test is run for. Another possibility is to perform a better calibration of the sensors before they are

used – we rushed our calibration slightly because we had our study participant coming at a specified time. We wanted to get all of our sensors calibrated before she came. We also moved our testing setup between calibration and testing, which may have led to inaccuracy in sensor calibration. We also noticed that as the test was running, the mean sensor values seemed to be increasing rather than remaining relatively constant, which may have resulted from a bug in the code or may have been from sensor saturation over a long time.

It is difficult to determine if our preliminary stationary pressure distribution results are similar to those from the Koo et al. experiment or not, as the Koo et al. research looked at absolute pressure values experienced in the foot at different slopes, and our sensor never was calibrated finely enough to do that accurately. However, we can roughly validate the accuracy and usefulness of our sensor with the aforementioned free-body-diagrams shown in Figures 1-3. As mentioned previously, the data that we obtained in our preliminary testing appears to match the results that we expect from our free body diagrams, so while we cannot conclusively validate the accuracy of our sensor because of the large confidence intervals in our data, the results seem promising.

5. Conclusion and Future Work

Through our data collection and analysis, it can be recommended that for generic shoes, increased arch support would benefit the user overall and help alleviate increased pressure found in the toe and heel when traveling up or downhill. Even slight increases in the arch support would benefit the consumer and allow for a more stable, comfortable shoe overall by directing more of the load to mid-foot. In the future, further validation of sensor accuracy would ensure more accurate data collection and, overall, a better understanding of pressure points throughout the foot on different inclines. Another future iteration would be to test the pressure distribution on a much wider range of inclines instead of just the 3 instances tested within this instance. More comprehensive data can also be collected by implementing dynamic testing to see pressure points instantaneously while the test subject is moving, as there is an expected increase in pressure throughout the toe, heel, and arch throughout a step. Lastly, an important future implementation would be to test different insoles with increased thicknesses in areas such as the arch to truly design the best insole for the specified consumer range.

References

- Che, H., Nigg, B. M., & de Koning, J. (1994). Relationship between plantar pressure distribution under the foot and insole comfort. *Clinical biomechanics* (Bristol, Avon), 9(6), 335–341.
[https://doi.org/10.1016/0268-0033\(94\)90062-0](https://doi.org/10.1016/0268-0033(94)90062-0)
- Cho, Y. J., Lee, D. W., Shin, H. S., Hwang, Y. B., Lee, D. O., Kim, D. Y., & Lee, D. Y. (2022). Change of In-Shoe Plantar Pressure According to Types of Shoes (Flat Shoes, Running Shoes, and High Heels). *Clinics in orthopedic surgery*, 14(2), 281–288. <https://doi.org/10.4055/cios20260>
- Hoffman, M. (2024, March 19). Foot Pain and Ankle Problems: Causes and Treatments. WebMD.
Retrieved May 2, 2024, from <https://www.webmd.com/pain-management/foot-pain-causes-and-treatments>
- Jordan, C., Payton, C., & Bartlett, R. (1997). Perceived comfort and pressure distribution in casual footwear. *Clinical biomechanics* (Bristol, Avon), 12(3), S5.
[https://doi.org/10.1016/s0268-0033\(97\)88312-x](https://doi.org/10.1016/s0268-0033(97)88312-x)
- Luuk, I., & guide, s. (n.d.). *50kg Load Cells with HX711 and Arduino. 4x, 2x, 1x Diagrams*. Circuit Journal. Retrieved May 2, 2024, from <https://circuitjournal.com/50kg-load-cells-with-HX711>

Appendices

Appendix A: Milestone Documentation

Milestone 1: Project Proposals

Ideas:

1. Distributed pressure on foot based on shoe type and sole

a. **Question:**

According to prior research ([NIH Paper 1](#), [NIH Paper 2](#), [NIH Paper 3](#), [The Foot Paper](#)) shoes are most comfortable when they cause the foot to experience the majority of pressure in the midfoot region. We want to investigate what insole shape and material stiffness will cause the majority of pressure to occur at the midfoot when a person is standing stationary.

b. **How we're going to answer the question:**

We will buy several different types of insoles off of [Amazon](#) and mount them on top of a variation of the pressure sensor shown in class. The insoles that we buy will be different shapes and we will have made sure that we can find data about their material stiffness online before we order them. We will measure the pressure in different parts of the foot in order to determine the pressure gradient along the foot and then compare the experimentally determined pressure gradients, insole shapes, and material stiffnesses for the different insoles in order to find if there is a correlation between those variables and if so, what combination of the shape and stiffness variables produce a gradient where the majority of pressure is in the midfoot region.

Because we are collecting data from commercial products, there are still many variables that we cannot perfectly control for when running these experiments. Therefore, after collecting the initial data from commercially produced insoles and analyzing that data we will create one or more of our own insoles that will be designed to confirm that using materials with the stiffness and shape that we recommend (with all other variables controlled as much as possible) is effective in causing the majority of foot pressure to occur at the midfoot.

c. **Data we're going to collect:**

We are going to collect pressure data on different parts of the bottom of the foot, including the toes, arch, and heel.

d. **How we're going to collect that data:**

We are going to collect this data by creating a universal sole. This sole will be composed of multiple force sensors located under each insole sample. Each sensor will be able to measure the force located in different locations on the foot.

e. **Previous research/Why the question is useful:**

The articles mentioned previously claim that the most comfortable insoles are those that apply the majority of pressure to the midfoot region. If we choose to measure the subjective 'comfort' level of each insole we could use that data to validate our data.

We hypothesize that insole shapes that fit to the foot better (i.e. have an arch) will apply more force to the midfoot region. We hypothesize the stiffness will not be directly correlated to where the pressure is applied on the foot but that there will be a high correlation between ‘comfort’ and low stiffness (lower stress to strain ratio). This question is useful because it can offer more insight into what materials and insole shapes provide the most foot comfort (and can be used to validate the performance of the pressure measuring tool from the previous year).

Milestone 2: Experimental Plan

Final Question: Is it more comfortable to walk uphill or downhill and what insole material stiffness and shape will produce more comfortable results for both of those stationary conditions?

- Standards of comfortability are defined by having most of the forces concentrated in the center of the foot. ([NIH Paper 1](#), [NIH Paper 2](#), [NIH Paper 3](#), [The Foot Paper](#))

Materials:

- Treadmill/Device to change elevation: -15 degrees to +15 degrees (\$0 using treadmills at Tufts Gym)
- A shoe with known material properties - will contact Converse to get data (and maybe a free pair of shoes??) - if not we can use one of our shoes
- [Force Sensor](#)
- 50kg sensor ships from China to US by March 10 - Total: \$15 for 8 sensors (shipping included)
- [Backup foot force sensor because we are ordering from China](#) (\$52)
- Insoles - will order later because we need to find a volunteer to help with testing and need shoe size (they will ship fast from Amazon) (~\$20)

Total Current Cost: \$35

Experimental Plan:

- Draw FBD of force on foot going uphill and downhill to show what we expect
- Build the sensor that has a foot insert using a previous model
- The “force-plate” that we build will be mounted inside the shoe that we purchase and we will place the insole that we are testing on top of the force plate inside of the shoe
- Build the software for the force-plate
- Calibrate the force-plate - make sure that the readings that we are getting from the plate are correct
- Place an object with a known weight on the force plate and verify that the data we get back from the force plate is correct

- One person will test each insole at specified inclines/declines (-15, -10, -5, 0, 5, 10, 15 deg) and we will use the sensor to generate data in LabView and analyze the data to see different pressure gradients based on the incline. We will use a heat map, 3D graph, etc to plot the data
- Through the collected data, we will design an insole that would be best to walk at an incline/decline.

Software:

- We will expand the software created by the previous lab group for one pressure measuring device to measure the pressure from all of the different devices in the “force-plate”

Challenges:

- Keeping foot position constant when testing
- Feet should be parallel with each other to equally distribute the force between both feet
- Validating data during the building phase
- Before we can start testing, we need to make sure the sensor actually works and provides correct results. By this, we mean that we limit the amount of noise from the testing data and run enough trials to “zero” the force plates.
- The sensors that we are interested in can measure up to 50kg. To ensure that we don’t overload the sensors, we need to find a participant for our study that is under this weight threshold.

Milestone 3 (*Background Section*):

In this study, we aim to study the pressure distribution from someone standing on an uphill and downhill incline. Using these findings, we will create an insole with a specified material stiffness and shape that will produce the most comfortable foot pressure gradient.

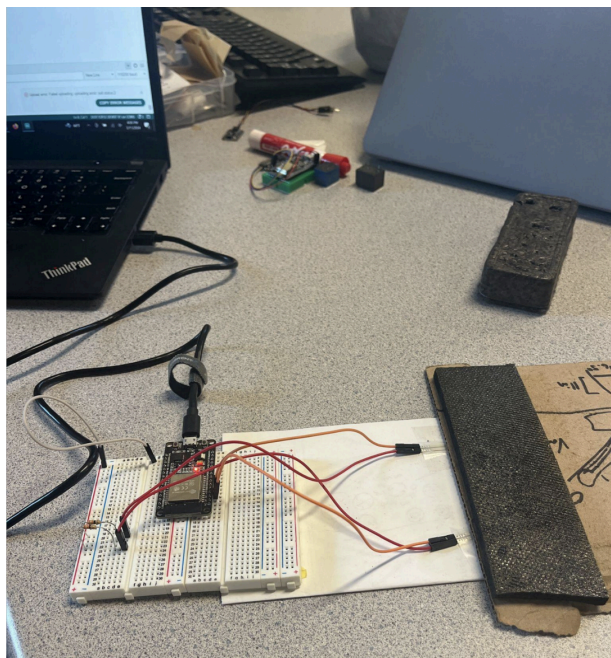
Our feet are essential to our everyday functioning, and they endure a lot of pressure from our bodies. When our feet are under uncomfortable conditions, it can lead to pain and biological problems for the feet. Feet pressure research has greatly contributed to studying how we can improve shoe comfort. In this study, we aim to contribute to foot pressure research in stationary conditions. Currently, a significant amount of data exists about foot pressures experienced on uphill and downhill slopes along with different walking/running conditions. However, there is little data on the foot pressures experienced within a stationary, inclined environment. Therefore, the data that we collect during the first stage of our research will add some new knowledge about foot pressure on an incline under stationary conditions.

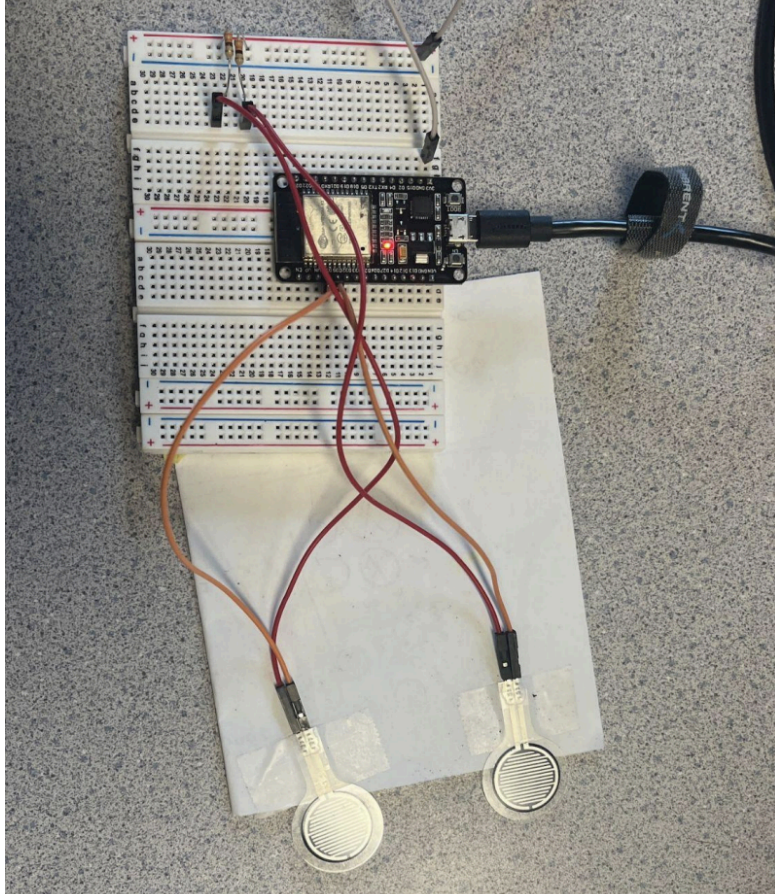
When characterizing the “comfort” of the foot under certain walking conditions, it is first important to define what “foot comfort” means. In a comfort study performed by Jordan et al, subjects walked 10m at their own pace and were asked to complete a perception of comfort questionnaire. The study found that most subjects reported that they were most comfortable when the pressure distribution was most concentrated in the midfoot region. Conversely, it was reported as less comfortable when the pressure distribution was more concentrated in the upper

and lower foot regions (Jordan et al). Based on the findings in this study, our study will set our standards of comfort to be when the pressure gradient is most concentrated in the midfoot region. Existing data on foot pressures experienced during uphill and downhill walking determined that “the peak pressure on the [medial midfoot] during downhill slope walking was lower than that during ground and uphill walking, whereas the peak pressure on the [medial midfoot] during uphill slope walking was similar to that during ground walking at each walking speed” (Koo et al). However, this study was performed on walking subjects rather than stationary subjects as is the case with our study, so while it may be useful to validate our data if the data differs it is not necessarily incorrect. Before we conduct the experiment, we plan to create free-body diagrams of the human foot that we will use to predict where the majority of foot pressures will be distributed. We will use these free-body diagrams in addition to the data from the aforementioned studies to validate our data.

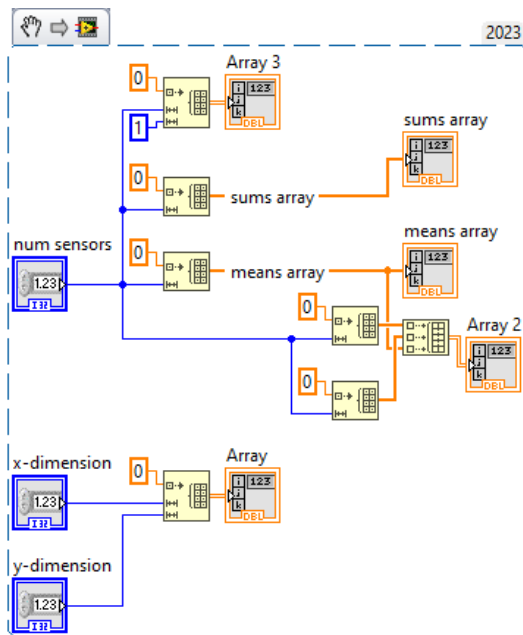
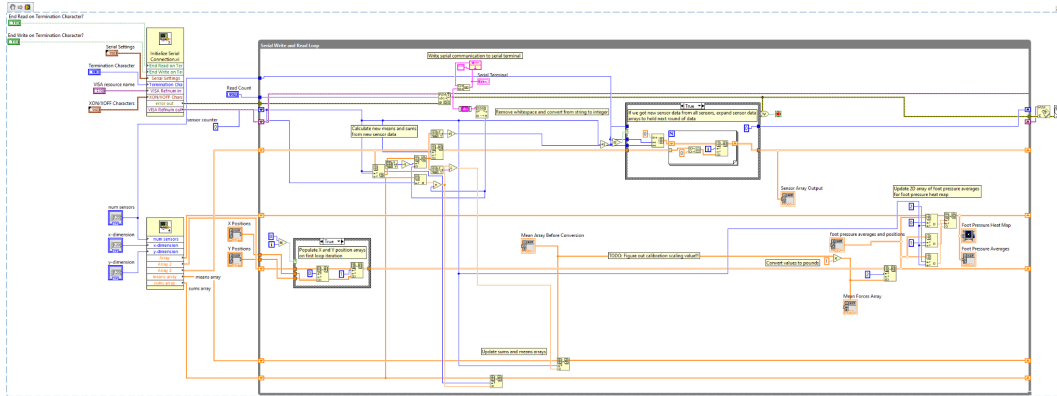
Our project will also add new knowledge to the field of insole design, specifically when designing for uphill and downhill conditions. A previous study found that when walking downhill, stiffer insoles with more arch support contributed to lower limb fatigue (Huang et al). However, we are going to look particularly at how adjusting the material that the insole is made of and the curvature of the insole distributes the pressure that the foot experiences and based on that data, make a recommendation on what materials should be used to construct the insole and what the shape of the insole should be. Our research, therefore, is different from previous research in that we will be approaching the problem from the perspective of how insoles can be improved rather than how well existing insoles help people with a certain foot shape.

Milestone 4: *Documentation of Constructed Experimental Setup*





Milestone 5: Software Development

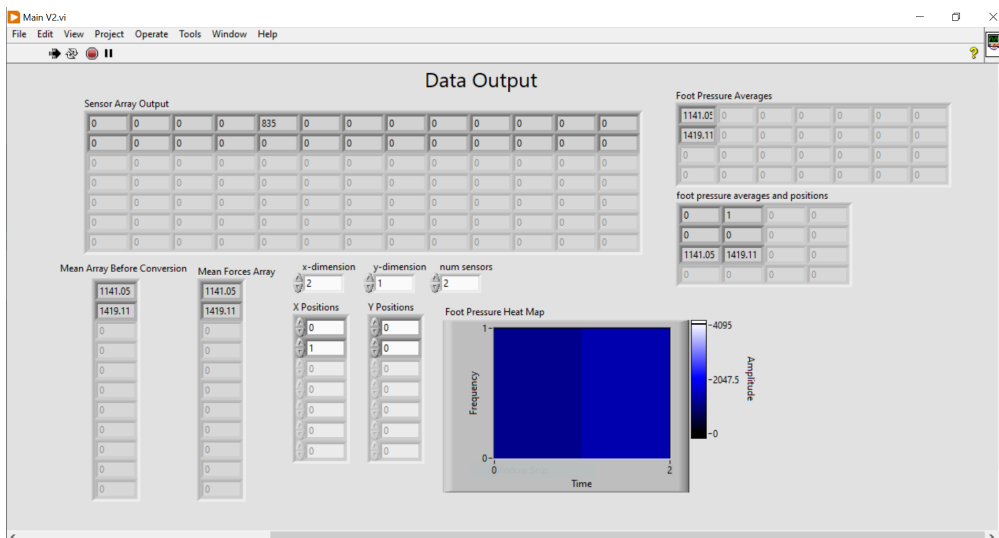


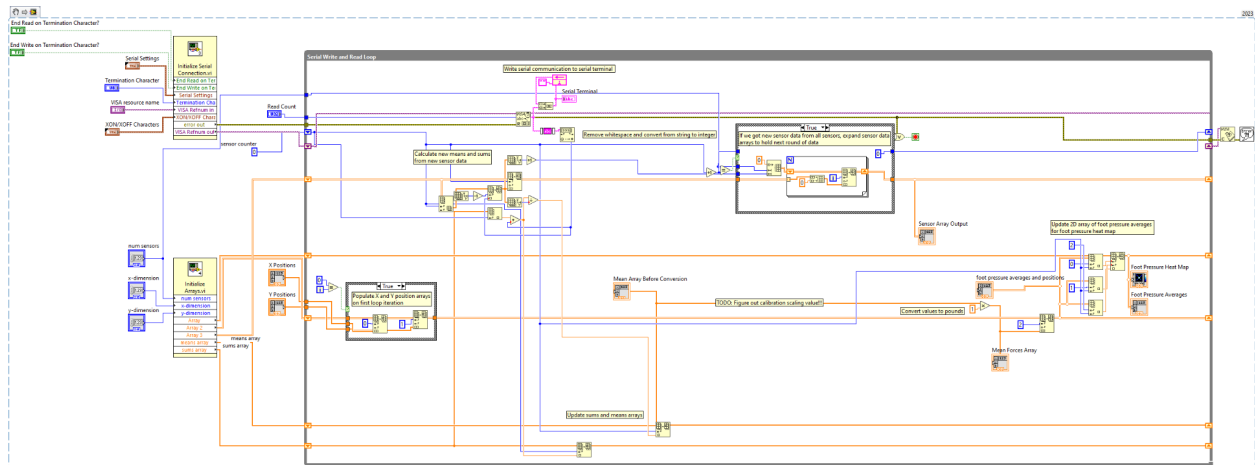
Final_Project_Arduino_Code_V1.ino

```

1  const int sens1 = 35;
2  const int sens2 = 34;
3
4  // variable for storing the potentiometer value
5  int sens1val = 0;
6  int sens2val = 0;
7
8  void setup() {
9      Serial.begin(115200);
10     delay(1000);
11 }
12
13 void loop() {
14     // Reading potentiometer value
15     sens1val = analogRead(sens1);
16     // Serial.print("1: ");
17     Serial.println(sens1val);
18
19     sens2val = analogRead(sens2);
20     // Serial.print("2: ");
21     Serial.println(sens2val);
22     delay(500);
23 }

```





End Read on Termination Character? End Write on Termination Character?

Serial Settings

baud rate: 115200

data bits: 8

parity: None

stop bits: 1.0

flow control: None

XON/XOFF Characters

XON: DC1

XOFF: DC3

VISA resource name: COM5

Termination Character: A

Read Count: 1000

Serial Terminal

```

0
0
0
0
n
0
0
0
0
835
n
  
```

Stop

Milestone 6: Foot Pressure Gradient Sensor Preliminary Data

Carolyn Thompson, Ethan Goldman, Madanna Vue, Tyler Mackowski

April 19, 2024

		x-dimension	y-dimension	num sensors			
		6	17	6			
X Positions	Y Positions	Raw Data Offsets	Calibration Slopes	Calibration Offsets			
3	2	0	-0.0026665	0.4224			
3	5	0	-0.0024537	-0.2634			
4	11	0	-0.002469	-0.4402			
2	17	0	-0.002469	-0.1933			
3	8	0	-0.0024537	-0.35			
1	13	0	-0.0023529	-0.2941			
3	0	0	0	0			

Figure 1. Sensor grid dimensions and calibration values for each sensor.



Figure 2. Experiment setup. Study participant stood on an uphill and downhill slope of pre-determined angle while data was taken.

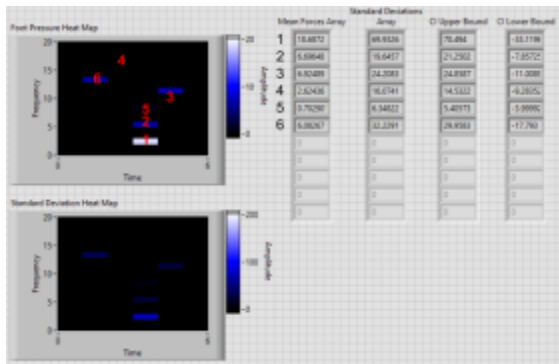


Figure 3. Foot pressure gradient on an uphill slope of 6.60°.



Figure 7. Uphill test setup.

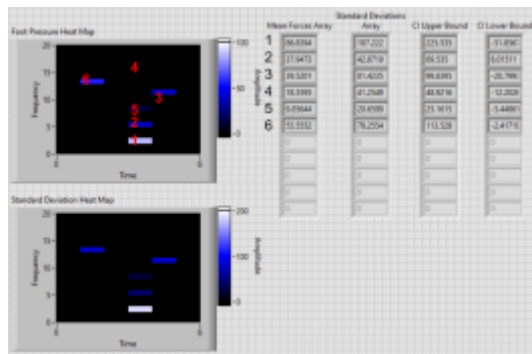


Figure 5. Foot pressure gradient on a downhill slope of 6.60°.



Figure 6. Downhill test setup.

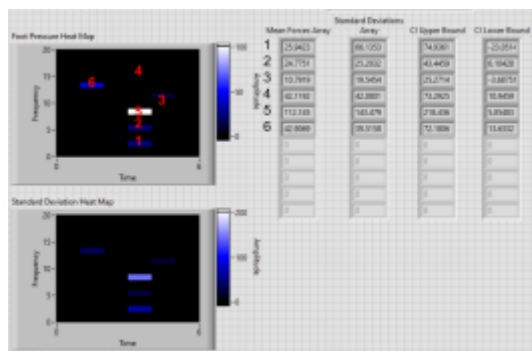


Figure 7. Foot pressure gradient on a flat surface.

Preliminary Analysis

For each of the uphill, downhill, and flat surface conditions we had the experiment participant stand on the foot pressure gradient sensor. Forces at each load cell in the sensor were collected with time, averaged, and then presented in the 'Foot Pressure Heat Map.' The standard deviation of the forces at each load cell over time were also calculated and presented in the 'Standard Deviation Heat Map'. Confidence intervals were calculated using the means, standard deviations, and with a 95% confidence interval.

We attempted to calibrate the load cells before collecting the data to convert from the load cell values to kilograms; however, when we began collecting the data the data was two magnitudes too small so we just increased the calibration slope values for each sensor by a magnitude of two from what it had been originally after calibration. Therefore, while the magnitudes of the load cells relative to each other are accurate, the actual magnitudes of the load cells are not guaranteed to be accurate.

A foam insert that was completely flat was used so the insert did not fit the participant's foot very well. We were concerned that the participant may have felt discomfort in the foot where the sensor was as the load cells that we used are made of large metal plates. We were concerned that the extra discomfort may have caused the participant to adjust their weight such that their weight was removed from the foot where the sensor was and thus we would not get the data from the participant's true standing position. However, when asked after the experiment if standing on the sensor had been uncomfortable the participant said that she hadn't noticed any discomfort at all!!s

The foot pressure heat maps that we obtained from the average data of each load cell actually make sense. For the uphill test, the majority of the weight was distributed to the heel, as expected, with some pressure at the front of the foot and very little in the middle. The downhill test again showed very little pressure in the midfoot region and surprisingly showed the most pressure again in the heel (it would be interesting to see if that continued to be the case in other trials because that was an unexpected result). However, the pressure at the front of the foot did increase and the pressure at the heel did decrease relative to the uphill experiment. The flat surface experiment showed the majority of the pressure distributed on the midfoot and big-toe of the participant with almost no pressure on the right-front part of the foot as opposed to the uphill and downhill cases. This change makes sense as when a person is standing on a flat surface their weight should be distributed more to the center of their foot.